Method for Walk Correction using Leading Edge Slope:

Derived from dual threshold feature of TOFHIR2C

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Abstract

Precision timing at the HL-LHC requires rapid, repeatable calibration as detector response evolves. With many interactions per bunch crossing, an external particle-arrival time reference is not readily available. The CMS barrel timing layer (BTL) therefore motivates calibration procedures based on internal measurements and a consistent parametrization across about 330 000 channels.

We extend an earlier dual-threshold bench study to BTL sensors and TOFHIR2C readout exposed to a 180 GeV pion beam. Two threshold-crossing times provide a leading-edge slope proxy. Its mean dependence on charge defines an inverse-slope walk correction with one amplitude parameter, AWC, per channel and configuration.

Comparison with independent microchannel-plate (MCP) reference calibrations shows good agreement between the correction shapes. Internal charge and slope information provide candidate predictors of AWC, with the MCP results serving as benchmarks. These results support development of a reference-free calibration procedure; predictive precision and transfer to other operating conditions require further validation.

BTL and TOFHIR2C: Refs. [3, 7]; dual-threshold motivation and bench study: Ref. [5].

Charge-based calibration works, but its coefficients vary

The MCP reference note obtains good walk descriptions with a quadratic in inverse charge [6]:

$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

Bar	End	a_i [ns]	b_i [ns ADC]	c_i [ns ADC^2]
7	Left	0.0796	238.3	-2.156×10^4
7	Right	-0.5155	422.6	-5.583×10^4
8	Left	0.1218	198.5	-1.523×10^4
8	Right	-0.2292	388.9	-5.132×10^4

The shape coefficients vary substantially even among four channels. Use pulse slope to seek a more consistent physical description.

Two bars in the reference module. Earlier time-of-flight calibration experience: ALICE [1, 2]. Q is the ADC charge proxy; a_i includes an arbitrary timing offset. Coefficient variation alone is not a measure of calibration precision.

TOFHIR2C measures the leading-edge slope with two thresholds

Larger pulses cross a fixed threshold earlier: the timestamp depends on amplitude [4]. Two-threshold timing has earlier implementations [8, 9]; TOFHIR2C provides both crossings [7, 5].

$$I_1 < I_2, \qquad t_1 < t_2, \qquad S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}].$$

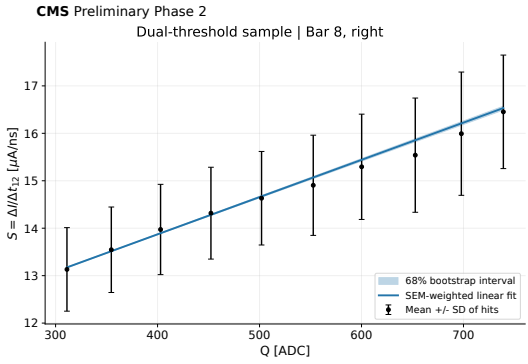
Locally, an approximately linear leading edge gives

$$\Delta t \simeq \frac{\Delta I}{S}.$$

Physical model: describe amplitude walk with an inverse slope, rather than an independent polynomial shape for every channel.

S is a finite-interval slope proxy, not the instantaneous derivative. The dual-threshold sample uses $\Delta I = 7.8125\,\mu\mathrm{A}$ from the implemented threshold conversion. The fitted AWC need not equal this threshold separation. GitHub: timing and threshold reconstruction [11].

Measure the charge-to-slope mapping using internal hits



Select both ends of one bar; leave the other module unconstrained. Use physical threshold crossings and the peak-right charge region. Fit ten profile means with SEM weights:

$$\overline{S}_i(Q) = c_{0,i} + c_{1,i} Q.$$

No MCP time enters this fit.

Reference module, bar 8, right end; dual-threshold sample: 19 312 selected hits. Points are mean slopes; bars show hit SD, not uncertainty on the mean. The band is the conditional 68% interval from 1000 hit-bootstrap refits with fixed bins. The profile averages the noisy single-hit slope measurements.

One amplitude describes walk once the slope is calibrated

$$W_i(Q) = \frac{A_i}{c_{0,i} + c_{1,i} Q}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W_i(Q).$$

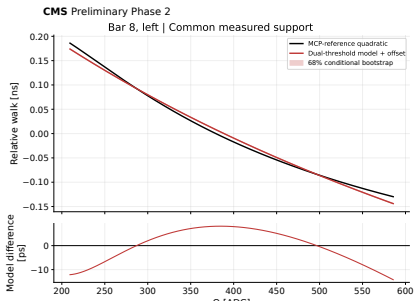
- Internal hits determine $c_{0,i}$ and $c_{1,i}$, the charge-to-slope mapping.
- One amplitude A_i (AWC, in μA) then sets the walk magnitude.
- AWC is constant per channel and configuration; the correction varies with Q.
- Channel timing offsets are a separate calibration.

The remaining task is to predict AWC from internal observables. Use the MCP-calibrated channels to develop and test that prediction.

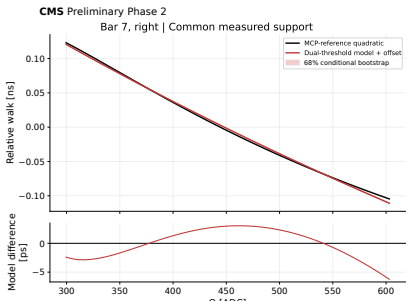
Apply the model within its calibrated charge range, with $c_{0,i} + c_{1,i} Q > 0$. The slope fit alone does not fix AWC. The common functional form follows the dual-threshold model in Ref. [5].

The single-amplitude model closely follows the MCP curves

Bar 8, left end

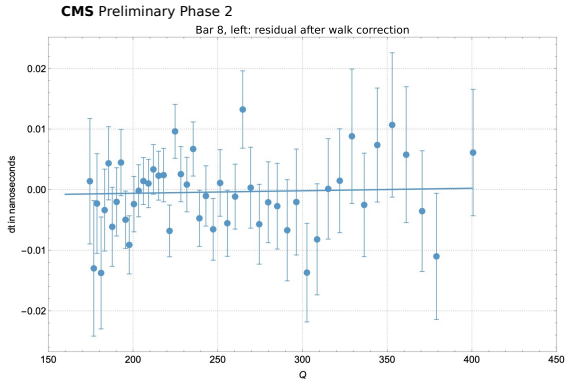


Bar 7, right end



Reference module. Black: MCP quadratic; red: dual-threshold model. Different-event samples; AWC and offset fitted only in common populated charge intervals. Bar 8, left end: 6.80 ps RMS curve mismatch. Lower panels show curve differences, not event resolution. Bands propagate slope-fit statistics with the MCP curve fixed. MCP clock alignment uses LOWESS [10]. This benchmarks shapes, not AWC prediction.

Residual charge dependence after the AWC correction



Bar 8, left end

Inspect the timing residual versus charge after the inverse-slope correction.

$0.01\,\mathrm{ns} = 10\,\mathrm{ps}$.

The residual linear trend is nearly flat over the displayed range.

This is not a single-hit timing-resolution measurement.

Original detector note, Figure 5: residual MCP-referenced timing, $t_{\mathrm{thr1}} - t_0$, after applying the AWC-type walk correction. Points and error bars are reproduced unchanged; the line is the original residual linear fit. This historical figure complements the curve-to-curve comparison on the preceding slide.

References

[1] "Performance of the ALICE Time-Of-Flight detector at the LHC", Eur. Phys. J. Plus (2013) 128: 44

[2] "PID performance of the ALICE-TOF detector at Run 2", N. Jacazio in Proceedings of LHCP2018, https://pos.sissa.it/321/232/.

[3] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

[4] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[5] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[6] S. White and Licheng J. Zhang, "September 2025 BTL Testbeam Reference Bar Walk Calibration from Hamamatsu MCP-PMT ",cms note.

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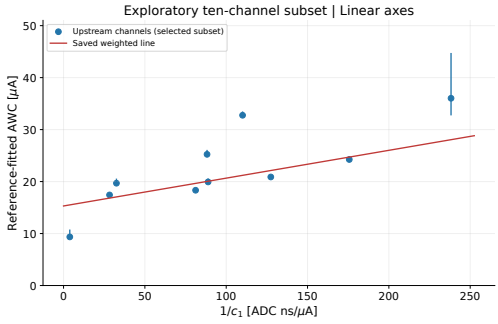
[8] J.E. Grund, SLAC-PUB-2416 (Oct. 1979).

[9] A.R. Frolov, T.V. Oslopova, Yu.N. Pestov, "Double threshold discriminator for timing measurements", Nuclear Instruments and Methods Volume 356, Issues 2/3 (1995) https://doi.org/10.1016/0168-9002(94)01239-3.

[10] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

[11] GitHub repository, https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector.

Backup: empirical AWC trend with inverse slope coefficient



The final observable of the original detector note: AWC versus $1/c_1$, the inverse slope-to-charge coefficient. Each point is one channel, not a charge bin. AWC is benchmarked against MCP; c_1 uses internal hits. The ten-channel trend motivates prediction, but the line is not a precision calibration ($\chi^2/\mathrm{ndf} = 538.6/8$). Bars: slope-only 16–84% bootstrap intervals; reference errors and correlations are omitted. More independent data are needed.